

[]: # 12412485

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[11]: import matplotlib.pyplot as plt
import math
import pandas as pd
from scipy.stats import norm
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[3]: def mean(arr):
    n = len(arr)
    ans = 0
    for num in arr:
        ans += num
    return ans/n

def variance(arr):
    n = len(arr)
    ans = 0
    x = mean(arr)
    for num in arr:
        ans += (num-x)**2
    return ans/n
```

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[4]: df = pd.read_csv("mark.csv")
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[5]: def normal(mean, var, x):
    a = 1/(((2*math.pi)**0.5) * var)
    b = 1/(2*(var))
    c = (x - mean)**2
    return a*(math.exp(-b*c))
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[6]: marks = df["Marks"]
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[7]: x_ = mean(marks)
var = variance(marks)
ndf = []
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[8]: for i in marks:
    ndf.append(normal(x_, var, i))
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[9]: plt.scatter(marks, ndf, color="blue", label="Data points")

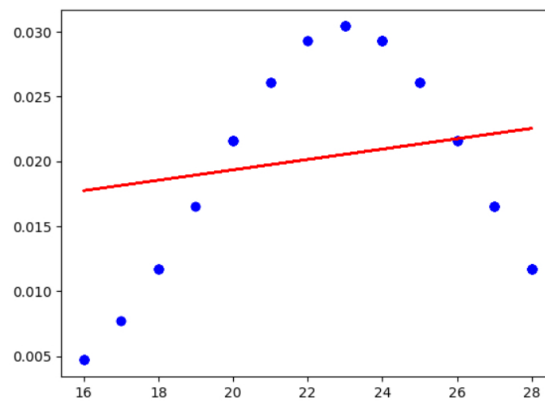
y_mean = mean(ndf)

num = sum((marks[i] - x_) * (ndf[i] - y_mean) for i in range(len(marks)))
den = sum((marks[i] - x_) ** 2 for i in range(len(marks)))
slope = num / den
intercept = y_mean - slope * x_

y_trend = [slope * xi + intercept for xi in marks]

plt.plot(marks, y_trend, color="red", label="Trend line")
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[9]: [<matplotlib.lines.Line2D at 0x29fedce02d0>]
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[12]: area = norm.cdf(1) - norm.cdf(-1)
print(area)

0.6826894921370859
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